Muneeb Lone
23i-2623
DS-B

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Assignment #6

1. $v = \begin{pmatrix} 4 \\ -3 \\ 7 \end{pmatrix}$ $A = \left[\begin{array}{ccc|c} 1 & 1 & -1 & 4 \\ 1 & -1 & 1 & -3 \\ 1 & -1 & -1 & 7 \end{array}\right]$

$$\left[\begin{array}{ccc|c} \textcircled{1} & 1 & -1 & 4 \\ 0 & -2 & 2 & -7 \\ 0 & -2 & 0 & 3 \end{array}\right] \begin{array}{l} \\ R2-R1 \\ R3-R1 \end{array}$$

$$\left[\begin{array}{ccc|c} \textcircled{1} & 1 & -1 & 4 \\ 0 & \textcircled{-2} & 2 & -7 \\ 0 & 0 & \textcircled{-2} & 10 \end{array}\right] \begin{array}{l} \\ \\ R3-R2 \end{array}$$

(i) $x_1 + x_2 - x_3 = 4$

(ii) $-2x_2 + 2x_3 = -7$

(iii) $-2x_3 = 10$

$$\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 1/2 \\ -3/2 \\ -5 \end{pmatrix} = \text{Coordinate Vector}$$

2. $[u]_B = \begin{pmatrix} 5 \\ -4 \\ 1 \end{pmatrix}$

$$u = B\,[u]_B = 5\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} - 4\begin{bmatrix} 1 \\ -1 \\ -1 \end{bmatrix} + \begin{bmatrix} -1 \\ 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 0 \\ 10 \\ 8 \end{bmatrix}$$ Ans.

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Q2: $C = \{1 + 2t + t^2, 3 - 3t, -1 + 5t^2\}$

1. $\left[\begin{array}{ccc|c} 1 & 3 & 0 & 0 \\ 2 & -3 & -1 & 0 \\ 1 & 0 & 5 & 0 \end{array}\right]$

$\left[\begin{array}{ccc|c} 1 & 3 & 0 & 0 \\ 0 & -9 & -1 & 0 \\ 0 & -3 & 5 & 0 \end{array}\right]$ $R2 - 2R1$, $R3 - R1$

$\left[\begin{array}{ccc|c} 1 & 3 & 0 & 0 \\ 0 & -9 & -1 & 0 \\ 0 & 0 & 5 & 0 \end{array}\right]$ $R3 + R2$

(C1, C2, C3)

- As there are pivots in every column, this system is linearly independent.
- Any element in P_2 can be written as a linear combination of the columns of this matrix hence it spans P_2 entirely as every row has a pivot.
- Hence C is a basis of P_2.

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2. $v = 2 + 3t - t^2$

Numerical

$$\left[\begin{array}{ccc|c} 1 & 3 & 0 & 2 \\ 2 & -3 & -1 & 3 \\ 1 & 0 & 5 & -1 \end{array}\right]$$

$$\left[\begin{array}{ccc|c} 1 & 3 & 0 & 2 \\ 0 & -9 & -1 & -1 \\ 0 & -3 & 5 & -3 \end{array}\right] \begin{array}{l} \\ R2-2R1 \\ R3-R1 \end{array}$$

$$\left[\begin{array}{ccc|c} 1 & 3 & 0 & 2 \\ 0 & -9 & -1 & -1 \\ 0 & 9 & -15 & 9 \end{array}\right] -R3 \times 3$$

$$\left[\begin{array}{ccc|c} 1 & 3 & 0 & 2 \\ 0 & -9 & -1 & -1 \\ 0 & 0 & -16 & 8 \end{array}\right] R3+R2$$

$$\left[\begin{array}{ccc|c} 1 & 3 & 0 & 2 \\ 0 & -9 & -1 & -1 \\ 0 & 0 & +2 & -1 \end{array}\right]$$

$x_3 = -\frac{1}{2}$

$x_2 = +\frac{1}{6}$

$x_1 = \frac{3}{2}$

$$[V_{dc} = \begin{pmatrix} 3/2 \\ 1/6 \\ -1/2 \end{pmatrix} = \frac{3}{2} + \frac{1}{6}t - \frac{1}{2}t^2$$

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3. $5\begin{pmatrix}1\\2\\1\end{pmatrix} + (-4)\begin{pmatrix}-3\\-3\\0\end{pmatrix} + 1\begin{pmatrix}0\\-1\\5\end{pmatrix}$

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$$u = \begin{pmatrix}-7\\21\\10\end{pmatrix}$$

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Q3: $W = \left\{ \begin{pmatrix} a & b \\ b & a \end{pmatrix} \mid a, b \in \mathbb{R} \right\}$ of $M_{2\times 2}$

1. $D = \left\{ \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \right\}$

$$c_1 \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + c_2 \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$$

$$\left[\begin{array}{cc|c} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{array}\right]$$

$$\left[\begin{array}{cc|c} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{array}\right] \begin{matrix} \\ \\ R3-R2 \\ R4-R1 \end{matrix}$$

- As this system has a pivot in every column we can deduce that D is linearly independant and the linear combinations of both matrices span the entirety of W. Hence this D is a basis of W.

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2. $E = \left\{ \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}, \begin{pmatrix} -2 & 1 \\ 1 & -2 \end{pmatrix} \right\}$

$$\left[\begin{array}{cc|c} \textcircled{1} & -2 & 0 \\ 2 & 1 & 0 \\ 2 & 1 & 0 \\ 1 & -2 & 0 \end{array}\right]$$

$$\left[\begin{array}{cc|c} \textcircled{1} & -2 & 0 \\ 0 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 0 \end{array}\right] \begin{array}{l} \\ R2-R3 \\ R3-2R1 \\ R4-R1 \end{array}$$

$$\left[\begin{array}{cc|c} \textcircled{1} & -2 & 0 \\ 0 & \textcircled{5} & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{array}\right] \text{Swap } R2, R3$$

As there is a pivot in every column, this system is linearly independent and hence the linear combinations of E can span the entirety of W. So E is a basis of W.

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3. $V = \begin{pmatrix} 5 & 3 \\ 3 & 5 \end{pmatrix}$

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w.r.t D:

$$\left[\begin{array}{cc|c} \textcircled{1} & 0 & 5 \\ 0 & 1 & 3 \\ 0 & 1 & 3 \\ 1 & 0 & 5 \end{array}\right]$$

$$\left[\begin{array}{cc|c} \textcircled{1} & 0 & 5 \\ 0 & \textcircled{1} & 3 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{array}\right] \begin{array}{l} \\ \\ R3-R2 \\ R4-R1 \end{array}$$

Coordinate Vector $= \begin{bmatrix} 5 \\ 3 \end{bmatrix}$

w.r.t E:

$$\left[\begin{array}{cc|c} \textcircled{1} & -2 & 5 \\ 2 & 1 & 3 \\ 2 & 1 & 3 \\ 1 & -2 & 5 \end{array}\right] \quad \text{Swap } R2, R3 \quad \left[\begin{array}{cc|c} \textcircled{1} & -2 & 5 \\ 0 & \textcircled{5} & -7 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{array}\right]$$

$$\left[\begin{array}{cc|c} \textcircled{1} & -2 & 5 \\ 0 & 0 & 0 \\ 0 & 5 & -7 \\ 0 & 0 & 0 \end{array}\right] \begin{array}{l} \\ R2-R3 \\ R3-2R1 \\ R4-R1 \end{array}$$

$[V]_E = \begin{pmatrix} 11/5 \\ -7/5 \end{pmatrix}$

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4. $U = 5\begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix} - 4\begin{pmatrix} -2 & 1 \\ 1 & -2 \end{pmatrix} = \begin{pmatrix} 13 & 6 \\ 6 & 13 \end{pmatrix}$ Ans.

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